

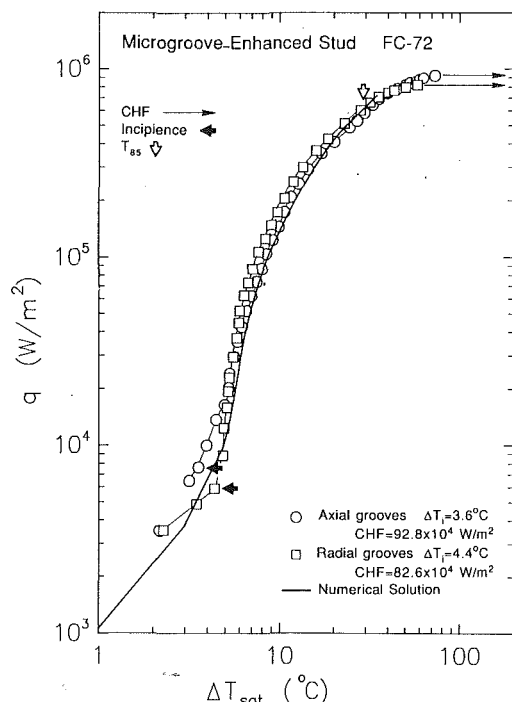


Fig. 12(a)

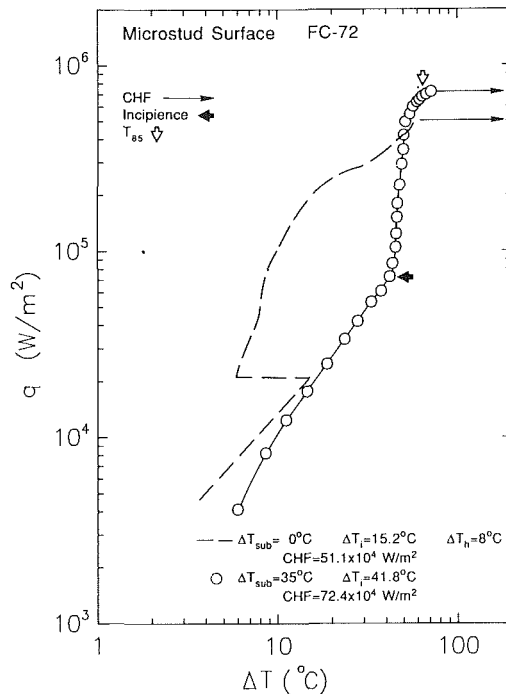


Fig. 13(a)

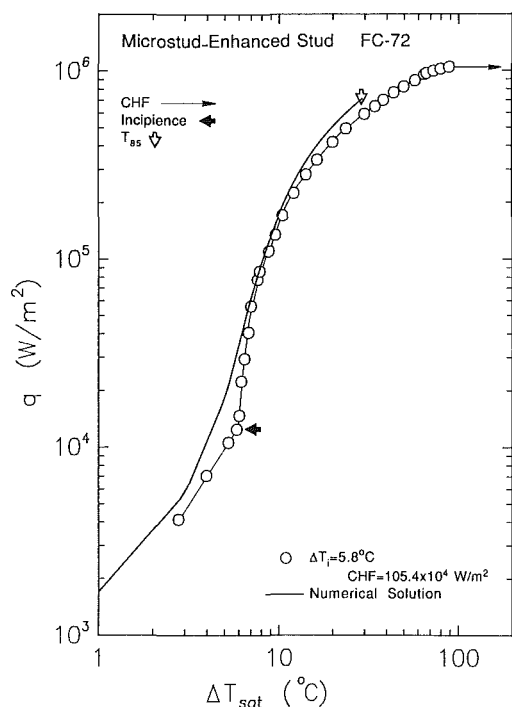


Fig. 12(b)

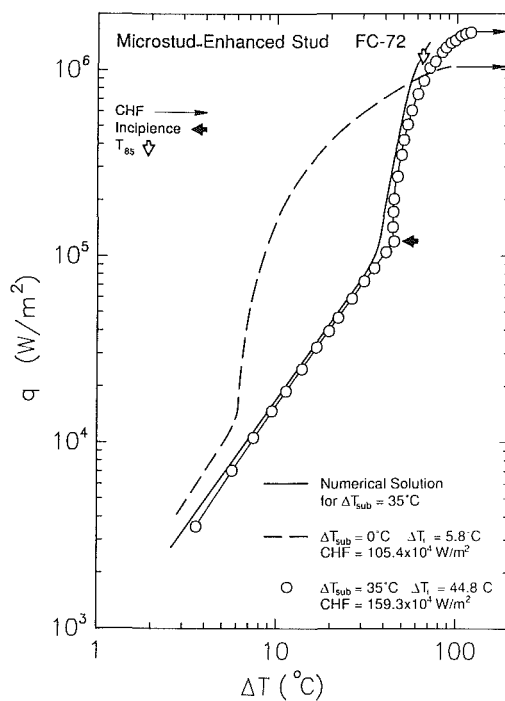


Fig. 13(b)

Fig. 12 Boiling curves for the (a) axially and radially grooved enhanced stud and (b) microstud-enhanced stud in saturated FC-72

Figure 14 shows a comparison of boiling behavior from the microstud-enhanced stud in saturated FC-72 and 35°C subcooled FC-72. Saturated boiling followed the general trend of boiling from a horizontal cylindrical surface. Nucleation commenced near the base with bubble departure diameters greater than the 0.305 mm microstud gap. Full activation of nucleate boiling over the entire perimeter and portions of the tip occurred at only 10.4 percent of CHF. Fully-developed boiling with severe vapor effusion was observed at approximately 46.5 percent of CHF. Following CHF, a thin vapor film formed

Fig. 13 Effect of subcooling on the boiling curves for (a) the low-profile microstud reference surface and (b) microstud-enhanced stud in FC-72

on the perimeter and tip regions, and vapor was released at the upper surface of the stud from one or two jets. Subcooled tests, Fig. 14, showed very noticeable differences in boiling behavior compared with saturated boiling. First, the bubbles were much smaller and emerged from within the microstud gaps. Second, most of the bubbles condensed in the subcooled pool just after departing the enhanced surface. Even during fully-developed boiling ($q = 28.6 \text{ W/cm}^2$), the large mass of vapor was almost totally condensed over a vertical distance equivalent to the stud diameter. Condensation at the vapor-